Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Network and Data Communications**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. Explain the concept of Binary Amplitude Shift Keying. Represent bit sequence 110010110 by the following waveforma. Manchesterb. Differential Manchester

2. Explain the design of Stop-and-Wait ARQ. Illustrate it with suitable flow diagram example.

3. What are the features of Link State routing protocols. Explain how Link State routing protocol can be used to find the shortest path with relevant example. What are its disadvantages.

4. Differentiate between switch, router, and hub.

5. What are the major differences between noise, distortion and attenuation?

6. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system considering all stations together produces 250 frames per second. What do you mean by vulnerable time of slotted ALOHA?

7. Why do we use NAT? Explain its different types.

8. Describe Quality of Service. What are its practical significances?

9. What is FTP and how does it work?

10. Explain Time Division Multiplexing with required figure.

11. What is MAC-address? The message sequence is 1101011011 and generator polynomial G(X) = x4+x+1. Calculate the transmitted encoded frame.

12. Write short notes on:a. Open-loopb. POP